

A DYNAMIC STRUCTURAL MODEL OF SUBSCRIPTION CHURN

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SETTING



Each month, a user chooses whether to continue subscribing to a music streaming platform.

Key Assumption: Platform Capital

The more hours a user accumulates on the platform, the more *platform capital* they have:

- E.g., curated playlists and algorithmic learning of music tastes
- More time spent $\Rightarrow$ higher utility from remaining subscribed

Research question: To what extent is consumer retention in digital subscriptions driven by platform capital versus platform frictions?

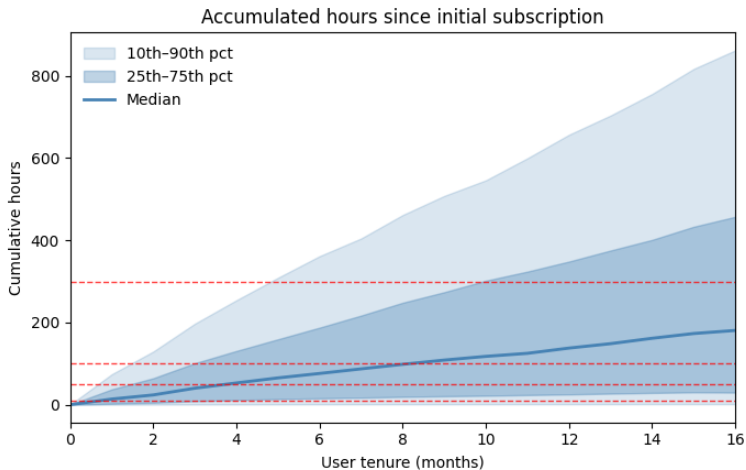
Source: KKBox (Kaggle)

User-month panel of subscription choices

- 9729 users, observed between Jan 2015 – Feb 2017
- 27.2% churn (average of 4.3 months subscribed before churn)
- 2.9% of users ever pause (average pause length is 2.8 months)
- *Eligibility*: registered within the observation window and made at least one transaction
- *Variables*: monthly hours played, auto-renewal status, subscription price

Notable Features

- Auto-renewal and price are *constant* within a user's spell
- Price varies *across* users (e.g., 99, 149, 180 TWD) but not within
- Paused users accumulate zero hours
- Churn is absorbing



MODEL

- Each 30-day period t after joining the platform, user i observes their state Ω_{it} and logit shocks $\{\varepsilon_{ita}\}_a$
- User chooses an action $a \in \{0, 1, 2\}$ (**Churn, Active, Pause**)
- State $\Omega_{it} = (h_{it}, a_{i,t-1})$ is a collection of accumulated hours h_{it} and an inertia term $\mathbb{1}\{a_{it} = a_{i,t-1}\}$
 - ▶ Discretized into $H = 5$ bins: 0–10, 10–50, 50–100, 100–300, 300+ hrs
- Latent-class structure addresses unobserved heterogeneity (e.g., *music lovers vs. casual users*)
- Law of motion for h_{it} :
 - ▶ *Active users*: hours accumulate, with hour bin transition probabilities estimated using empirical frequencies
 - ▶ *Paused users*: **do not accumulate hours** (zero probability of transitioning between bins) and **decay probability** $\rho = 0.1$ of reentering lower state
 - ▶ *Churned users*: enter an **absorbing terminal state** with **0 utility** for all future periods

UTILITY SPECIFICATION

Action-Specific Value Functions

The expected discounted utility of choosing action a and behaving optimally thereafter is $V_a(\Omega_{it}) = \bar{V}_a(\Omega_{it}) + \varepsilon_{ita}$, where

$$\bar{V}_0(\Omega_{it}) = 0$$

$$\begin{aligned}\bar{V}_1(\Omega_{it}) = & \bar{u}_1 + \alpha R_{it} - \beta p_{it} + \eta \mathbb{1}\{a_{i,t-1} = 1\} + \sum_{k=2}^H \gamma_k \mathbb{1}\{h_{it} = k\} \\ & + \delta \mathbb{E}[\bar{V}(\Omega_{i,t+1}) \mid \Omega_{it}, a = 1]\end{aligned}$$

$$\begin{aligned}\bar{V}_2(\Omega_{it}) = & \bar{u}_2 + \eta \mathbb{1}\{a_{i,t-1} = 2\} \\ & + \delta \mathbb{E}[\bar{V}(\Omega_{i,t+1}) \mid \Omega_{it}, a = 2]\end{aligned}$$

The Bellman equation is

$$V(\Omega_{it}) = \max_a \{V_a(\Omega_{it})\}$$

ESTIMATION RESULTS

Parameter	Estimate	S.E.	95% C.I.
α	1.012	0.054	[0.906, 1.118]
β	-0.029	0.001	[-0.030, -0.027]
η	1.637	0.072	[1.495, 1.779]
γ_2	-0.505	0.136	[-0.773, -0.238]
γ_3	-0.683	0.211	[-1.097, -0.269]
γ_4	-0.541	0.216	[-0.965, -0.117]
γ_5	-0.205	0.187	[-0.571, 0.162]
<i>Class 1 (89.4%)</i>			
$\bar{u}_1$	5.055	0.240	[4.584, 5.526]
$\bar{u}_2$	-1.506	0.234	[-1.964, -1.047]
<i>Class 2 (10.6%)</i>			
$\bar{u}_1$	5.643	0.219	[5.214, 6.073]
$\bar{u}_2$	0.903	0.135	[0.638, 1.168]

Notes: Robust standard errors clustered at the user level. $\delta = 0.99$, $\rho = 0.10$, $LL = -12,550.9$.

Homogeneous model

Heterogeneity on γ

COUNTERFACTUALS

How do the choices of a subscription platform designer affect CLV, retention, and churn? Consider...

❶ Eliminating auto-renewal

- ▶ *Model:* Turn auto-renewal indicator $R_{it} = 0$ for all users.

❷ Increasing cancellation frictions

- ▶ *Model:* Increasing the inertia parameter η only on currently active users.

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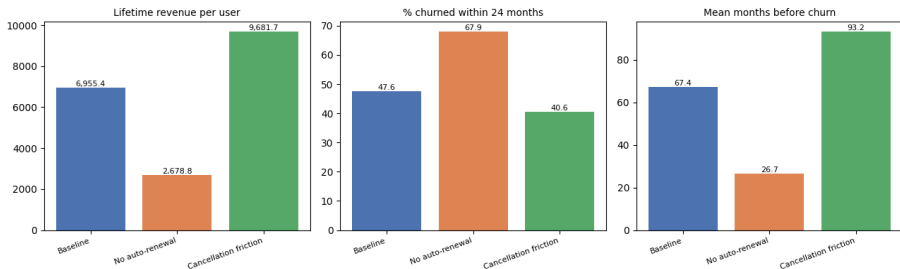
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2 Increasing cancellation frictions

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APPENDIX: HOMOGENEOUS MODEL

Parameter	Estimate	S.E.	95% C.I.
$\bar{u}_1$	4.298	0.242	[3.824, 4.772]
$\bar{u}_2$	-0.765	0.050	[-0.863, -0.666]
α	1.129	0.060	[1.011, 1.247]
β	-0.025	0.001	[-0.027, -0.023]
η	2.254	0.051	[2.154, 2.354]
γ_2	-0.535	0.153	[-0.834, -0.236]
γ_3	-0.009	0.165	[-0.332, 0.314]
γ_4	-0.237	0.128	[-0.488, 0.015]
γ_5	-0.722	0.131	[-0.978, -0.465]

Notes: Robust standard errors clustered at the user level. $\delta = 0.99$, $\rho = 0.10$, $LL = -12,690.8$.

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APPENDIX: HETEROGENEITY ON γ

Parameter	Estimate	S.E.	95% C.I.
α	0.940	0.055	[0.831, 1.048]
β	-0.029	0.001	[-0.030, -0.027]
η	1.602	0.041	[1.521, 1.683]
<i>Class 1 (82.2%)</i>			
$\bar{u}_1$	1.011	1.281	[-1.501, 3.522]
$\bar{u}_2$	-2.385	0.281	[-2.935, -1.834]
γ_2	-4.281	2.283	[-8.756, 0.194]
γ_3	-4.493	3.582	[-11.513, 2.527]
γ_4	-0.350	1.875	[-4.025, 3.324]
γ_5	4.662	1.272	[2.168, 7.155]
<i>Class 2 (17.8%)</i>			
$\bar{u}_1$	6.230	0.238	[5.763, 6.697]
$\bar{u}_2$	0.422	0.139	[0.148, 0.695]
γ_2	-0.656	0.155	[-0.959, -0.353]
γ_3	-0.423	0.195	[-0.806, -0.041]
γ_4	-0.984	0.155	[-1.287, -0.681]
γ_5	-1.114	0.178	[-1.463, -0.764]

Notes: Robust standard errors clustered at the user level. $\delta = 0.99$, $\rho = 0.10$, $LL = -12,430.7$.